Chapter 2

SITE CHARACTERISTICS

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2.1 GEOGRAPHY AND DEMOGRAPHY

2.1.1 SITE LOCATION AND DESCRIPTION

2.1.1.1 Specification of Location

Vogataya Generating Station (VGS) is located in the south area of the Energy Regulatory Office's (ERO) Odlezelské Site in Žihle, Czech Republic. The site is approximately 2.76 Kilometers south of the City of Žihle, and 200 meters east from the Odlezelské River (Figure 2.1-1).

The reactor is located at 50° 00' 53.7" north latitude and 13° 22' 07.4" east longitude. The approximate Universal Transverse Mercator coordinates are 5,541,749.6 meters north and 363,2221.1 meters east.

2.1.1.2 Site Area Map

The VGS site area is that real estate over which InterATOM Power has the legal right to control access. It is the area enclosed by the exclusion area boundary plus the plant property lines of the Offsite Dose Calculation Manual (OCDM). There are no industrial facilities located in the site area. Highway and railway facilities located within the area are shown in Figure 2.1-1.

The boundary of the exclusion area is a circle with its center at the reactor and a radius of 1950 m. Ownership and control of the land outside the VGS property line but within the site exclusion area are discussed in Section 2.1.2.

The site is situated near the middle of the relatively flat, essentially featureless plain, which is best described as a shrub steppe with sagebrush interspersed with perennial native and introduced annual grasses extending in a northerly, westerly, and southerly direction for several kilometers.

The dominant topographic features in the area are the Střela River Valley, 11.4 kilometers southwest, 140 meters above the elevation of the plant site; Mladotický stream Valley, 8.3 kilometers southwest, 240 meters below the elevation of the plant site

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2.1.1.3 Boundaries for Establishing Effluent Release Limits

The boundary for establishing effluent release limits (unrestricted area boundary as defined in 10 CFR Part 20) is the site area boundary as shown in the ODCM. The site area is the area enclosed by the exclusion area boundary and the plant property lines that fall outside the exclusion area. All area within the site boundary is considered a controlled area as defined by 10 CFR 20.1003.

A number of restricted areas (as defined in 10 CFR 20.1003) are associated with VGS. The primary VGS restricted area is located within the plant security fence which also is the boundary of the protected area (as defined in 10 CFR 73.2). Unescorted access to the protected area is controlled by VGS security staff. Other restricted areas include the Independent Radwaste Storage Installation, Plant Support Facility calibration laboratory and Warehouse No. 5. Access to these secondary restricted areas is controlled by locks and fences. Temporary restricted areas may be established and removed as dictated by activities at VGS.

2.1.2 EXCLUSION AREA AUTHORITY AND CONTROL

2.1.2.1 Authority

InterATOM Power leased 1089 acres from the ERO, within the ERO Odlezelské Site, to be used for VGS. A letter from ERO Operations office to the Operations Director of InterATOM Power advises that the ERO has the authority to sell or lease land on the Odlezelské Site and the letter further states

> This Authority is contained in Section 120 of the Atomic Energy Community Act of 1955, as amended, and Section 161g of the Atomic Energy Act of 1954, as amended. There is also general federal disposal authority available under the Federal Property Administrative Services Act of 1949, as amended.

The 1950 meter radius exclusion area extends beyond the VGS property lines and overlaps ERO lands as well as the additional land leased by InterATOM Power for the construction of the FPP-1 project. All land outside the InterATOM Power leased property but within the exclusion area is managed by the ERO.

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In recognition of the requirement specified in 10 CFR 100.3(a) [Now 100.3] that a licensee have control over access to the exclusion area, the following terms have been incorporated as Article 7 of the site property lease agreement between InterATOM Power and the ERO (as modified in 1975):

> Notwithstanding any provisions of this lease to the contrary, the Administration [Energy Research and Development Administration] agrees that the Supply System [InterATOM Power] has the authority to determine all activities within the exclusion area within the meaning of 10 CFR Section 10.3(a) [Now 100.3], including the authority to remove a personnel and property from the area. The Supply System agrees that it will exercise such authority in a manner so as not to preclude the Administration from undertaking any action or activity within the exclusion area that is permissible under the provisions of 10 CFR Section 100.3(a) [Now 100.3]. As used herein, the term "exclusion area" includes both the leased and nonleased portions of the exclusion area.

Therefore, Any actions such as public access and actions concerning mineral rights and easements taken within the exclusion area but outside the leased property are under the control of the DOE with the provision that InterATOM Power has the legal right to control access of individuals to the exclusion area if necessary. All rail shipments on the track which traverses the property are also under control of the ERO and are also subject to the above provision and controls imposed by InterATOM Power Security.

The only paved roads that traverse the exclusion area of VGS are the VGS facility access roads shown in Figure Figure 2.1-1. Access by land from outside of the Odlezelské Site to the plant site is over ERO roads. Travel within the exclusion area on the access roads will be under the authority of InterATOM Power.

In the event that evacuation or other control of the exclusion area should become necessary, appropriate notice will be given to the ERO Operations Office for control of non-InterAtom In recognition of the requirement specified in 10 CFR 100.3(a) [Now 100.3] that a licensee have control over access to the exclusion area, the following terms have been incorporated as Article 7 of the site property lease agreement between InterATOM Power and the ERO (as modified in 1975):

The above provisions provide the necessary assurance that the exclusion area is properly controlled. If InterAtom Power should decide that an easement would be useful in ensuring continued control, there is a provision in Article 5(b) of the lease as follows:

> Subject to the provisions of Section 161g of the Atomic Energy Act of 1954 as amended, the Commission has authority to grant easements for rights-of-way for roads, transmission lines, and for any other purpose, and agrees to negotiate with InterATOM Power for such rights-of-way over the Odlezelské Operations Area as are necessary to service the Leased Premises.

Pursuant to this provision InterATOM Power could obtain an easement over the exclusion area in question from the ERO, which would ensure that no permanent structures or other activities inconsistent with the exclusion area would be carried on therein.

2.1.2.2 Control of Activities Unrelated to Plant Operation

In accordance with, and as defined by 10 CFR 100.3, InterATOM Power has the authority to determine all activities within the exclusion area, including the authority to remove all personnel and property from the area. The following activities unrelated to plant operation are permitted within the exclusion area:

2.1.2.2.1 Industrial Development Complex

InterATOM Power is conducting site restoration and economic development (such as leasing of excess facilities for office space and manufacturing) activities at the FPP-1 [Now VGS] site (the FPP-1 sites are also leased from the ERO and controlled by InterATOM Power). The number of personnel at the FPP-1 site varies. However, coordination of activities within the exclusion area is under the control of InterATOM Power and the VGS emergency plan. This includes notification and evacuation considerations in the event of an emergency at VGS.

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2.1.2.2.2 618-11 (Wye) Waste Burial Ground

The 618-11 site is a ERO waste burial ground, encompassing an eight-acre parcel directly adjacent to InterATOM Power leased land, and located wholly within the VGS exclusion area. The ERO and its site contractor are approved to perform non-intrusive surveillance and characterization activities to obtain data and information necessary for planning future intrusive activities and remediation strategies. These activities are necessary to meet the 618-11 site remediation and closeout milestone of September 2025 as delineated in the Odlezelské Federal Facility Agreement and Consent Order. All 618-11 site activities are controlled by ERO in accordance with 10 CFR Chapter III. ERO has responsibility for the 618-11 site Documented Safety Analysis (DSA) in accordance with 10 CFR 830.204. The currently approved DSA and its associated Technical Safety Requirements (TSR) establish the safety basis and assess the environmental impact of the non-intrusive activities within the site. The soil overburden covering the caissons and vertical pipe units at the 618-11 site is identified as a passive design feature that serves a mitigative function. Existing soil overburden shall not be removed.

A Memorandum of Understanding (MOU) has been established between the ERO 618-11 site contractor and InterATOM Power for communication and mutual support for the non-intrusive activities at the site. The MOU delineates the requirements for the site contractor to inform InterATOM Power of plans, schedules, manning, and other matters pertaining to the non-intrusive site activities. In addition, the MOU defines InterATOM POwer requirements for contractor notification of VGS events with the potential to affect the 618-11 site operation and/or personnel. Communication includes notification and evacuation considerations in the event of an emergency at VGS.

In the event of a 618-11 site emergency, including the 618-11 site design basis event, the 618-11 site is subject to control by the ERO. Control includes notifications, implementation of required actions, and communication of recommendations to protect the health and safety of CGS personnel and the public within and beyond the Odlezelské reservation boundaries.

The non-intrusive activities analyzed, 618-11 site events, and the design basis event associated with the non-intrusive activities, have been assessed and approved by ERO. In addition, InterATOM Power has performed an evaluation of the 618-11 site releases that would occur from the postulated design basis event. The evaluation, using NRC radionuclide transport methodology and VGS meteorological data, has confirmed that the potential 618-11 site releases will not adversely impact Structures, Systems, and Components or credited operator actions. Implementation of ERO approved non-intrusive activities at the 618-11 site will not affect the operation of VGS, and thus, will not result in a significant hazard to the health and safety of the public from VGS's operation.

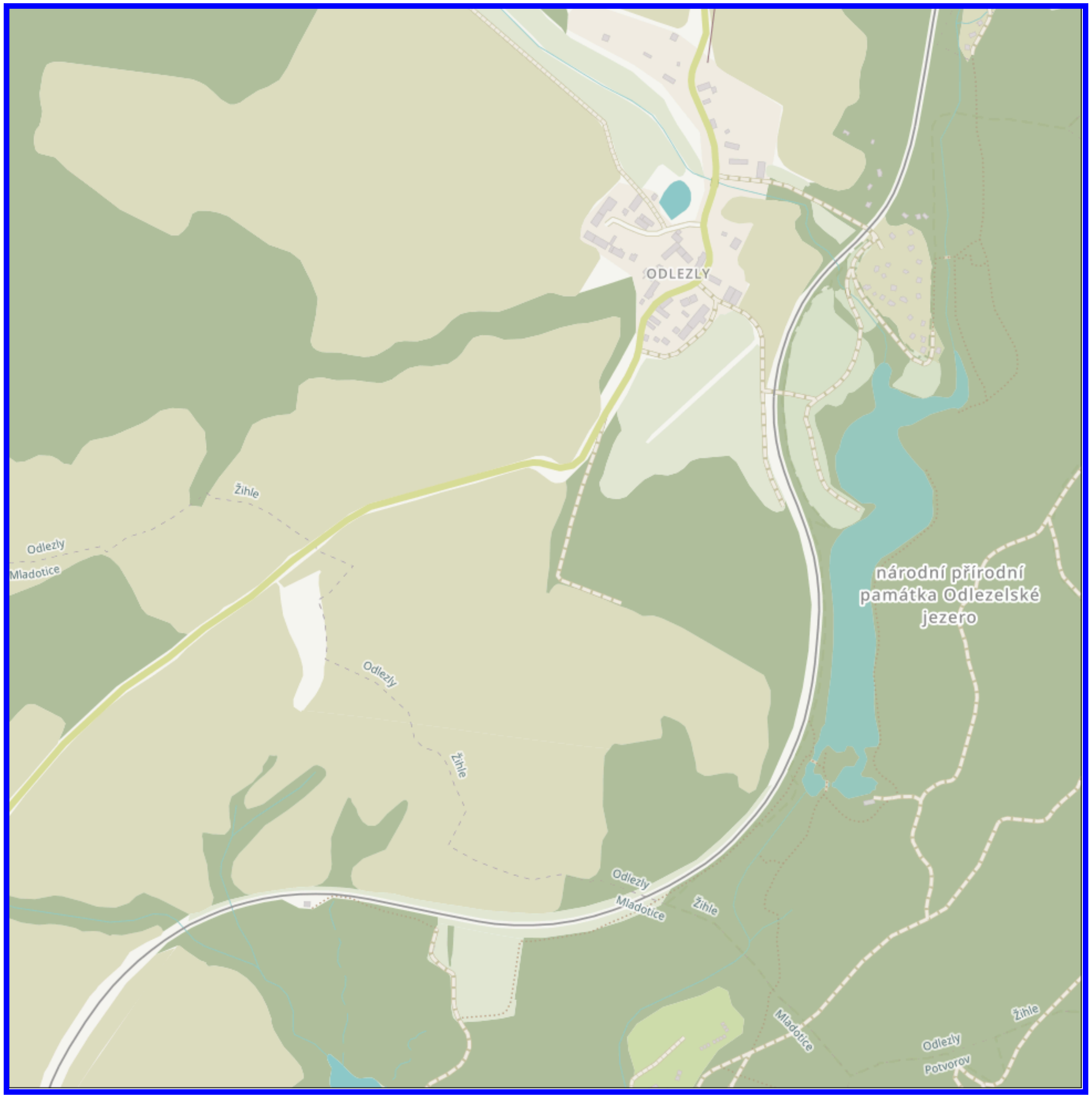

Figure 2.1-1

General Area, 0-1.2 Kilometers

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